

Unit 10, Lesson 8: Exploring Functions Using Equations, Tables, and Graphs

Benchmarks

MA.8.F.1.2 Given a function defined by a graph or an equation, determine whether the function is a linear function. Given an input-output table, determine whether it could represent a linear function.

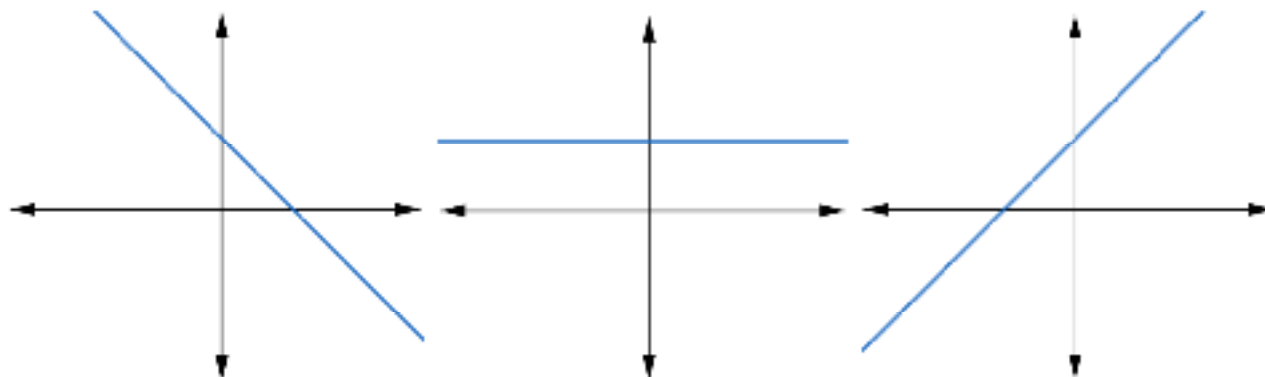
MA.8.AR.2.3 Given an equation in the form of $x^2 = p$ and $x^3 = q$, where p is a whole number and q is an integer, determine the real solutions.

Learning Targets

- ☐ I can create a table of values from the equation of a function.
- ☐ I can graph a function using a table of values.

10.8.1 Warm-Up: Bigger and Smaller

Three graphs are shown.



Diego said, "The graphs are ordered from smallest to largest." Maya disagreed and said, "No. They are ordered from largest to smallest."

But these are graphs, not numbers!

1. What do you think Diego and Maya are thinking?

10.8.2 Collaboration: Revisiting Squares and Cubes

Work with your partner on the following.

1. Solve each equation for x . Show your work.

$x^2 = 16$

$x^2 = 1$

$x^2 = 36$

$x^2 = 100$

$x^2 = 0$

2. Use your solutions to the previous equations to complete a table of values for the equation $y = x^2$.

x									
y									

3. Solve each equation for x . Show your work.

$x^3 = 8$

$x^3 = -64$

$x^3 = -1$

$x^3 = 125$

$x^3 = 0$

$x^3 = 64$

$x^3 = -125$

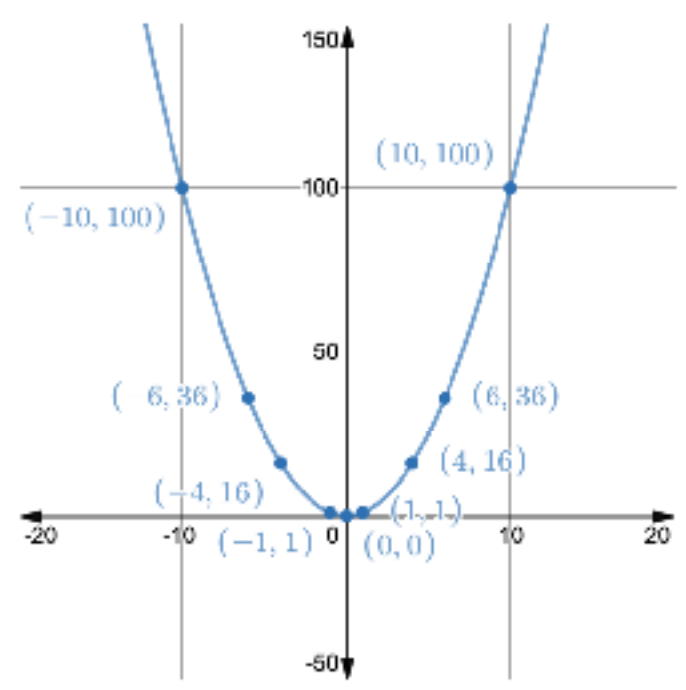
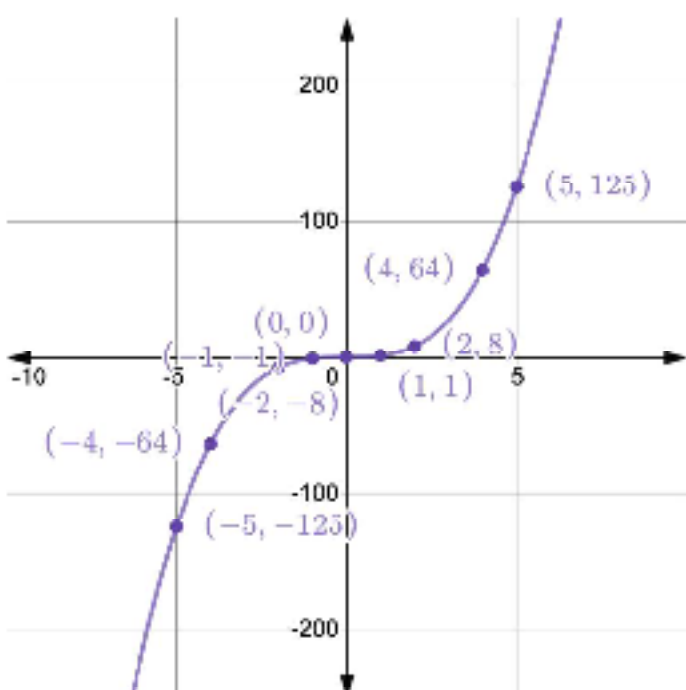
$x^3 = -8$

$x^3 = 1$

4. Use your solutions to the previous equations to complete a table of values for the equation $y = x^3$.

x									
y									

5. The graphs of both equations are shown. Write the equation that matches each graph based on the tables of values created in the previous problems.

Equation	Graph
	
	

10.8.3 Exploration: Equations and Their Graphs

Relations between two variables can be represented using equations, tables of values, and graphs.

- 1. For each of the equations given, work with your partner to create a table of values. Then, plot each ordered pair from the table of values, using them to create a rough sketch of the graph by connecting the points.

One of the graphs and its corresponding table from the collaboration activity have already been included.

Equation 1	Graph												
$y = 2x - 7$													
Table of Values													
<table border="1"><thead><tr><th>x</th><th>y</th></tr></thead><tbody><tr><td>-3</td><td></td></tr><tr><td>-1</td><td></td></tr><tr><td>0</td><td></td></tr><tr><td>2</td><td></td></tr><tr><td>6</td><td></td></tr></tbody></table>		x	y	-3		-1		0		2		6	
x		y											
-3													
-1													
0													
2													
6													

Equation 2

$y = x^2 + 2$

Table of Values

x	y
-2	
-1	
0	
1	
2	

Graph

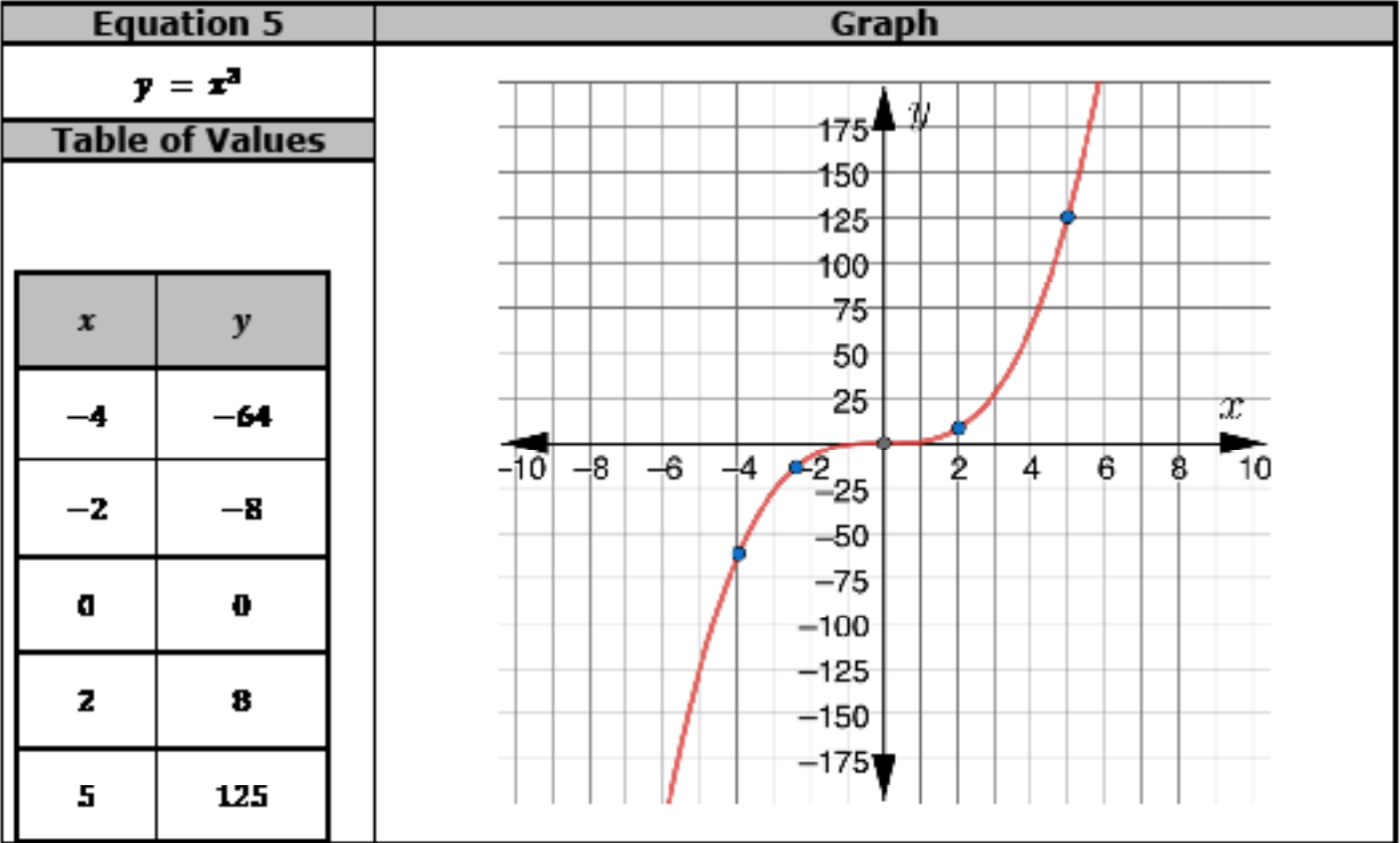
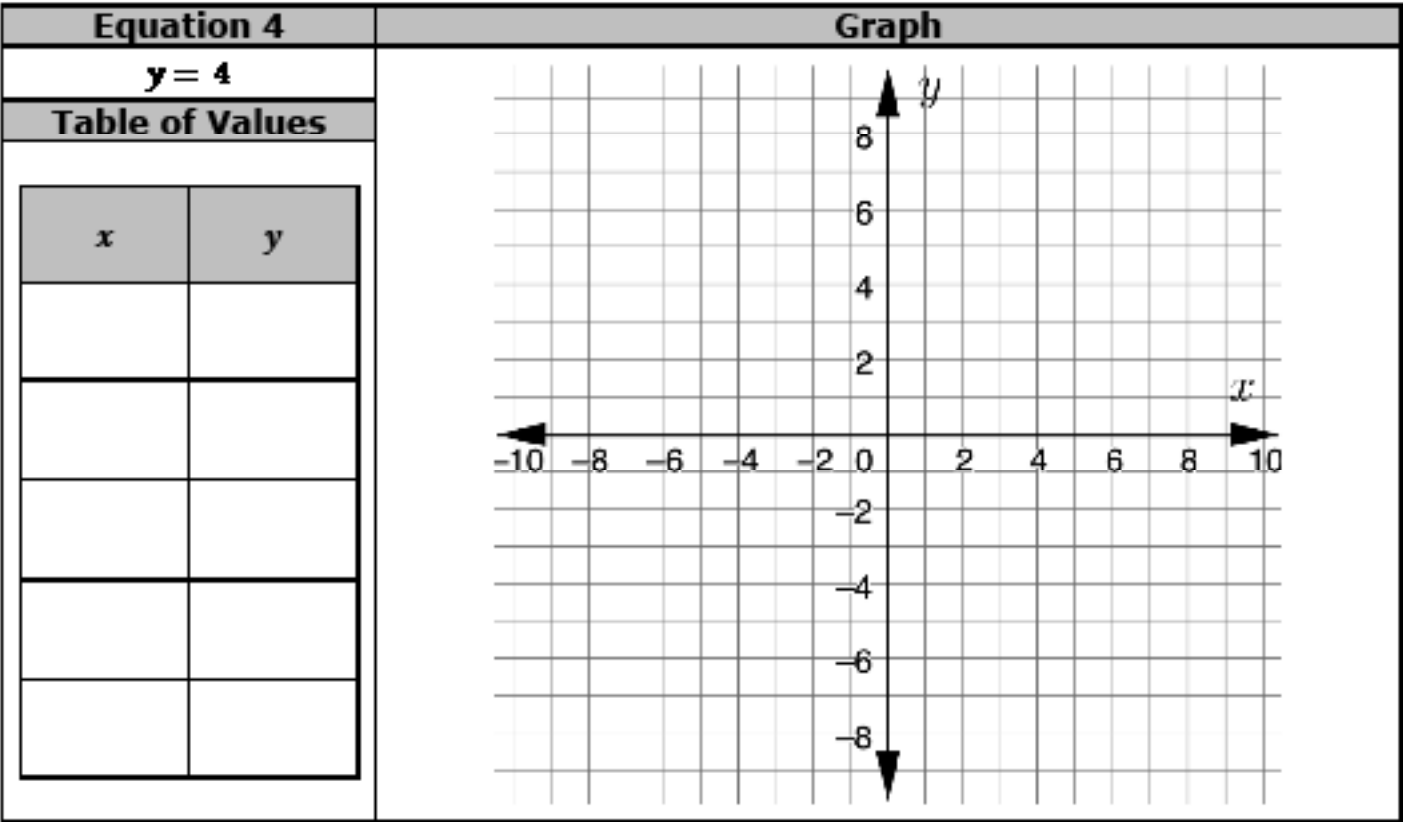
Equation 3

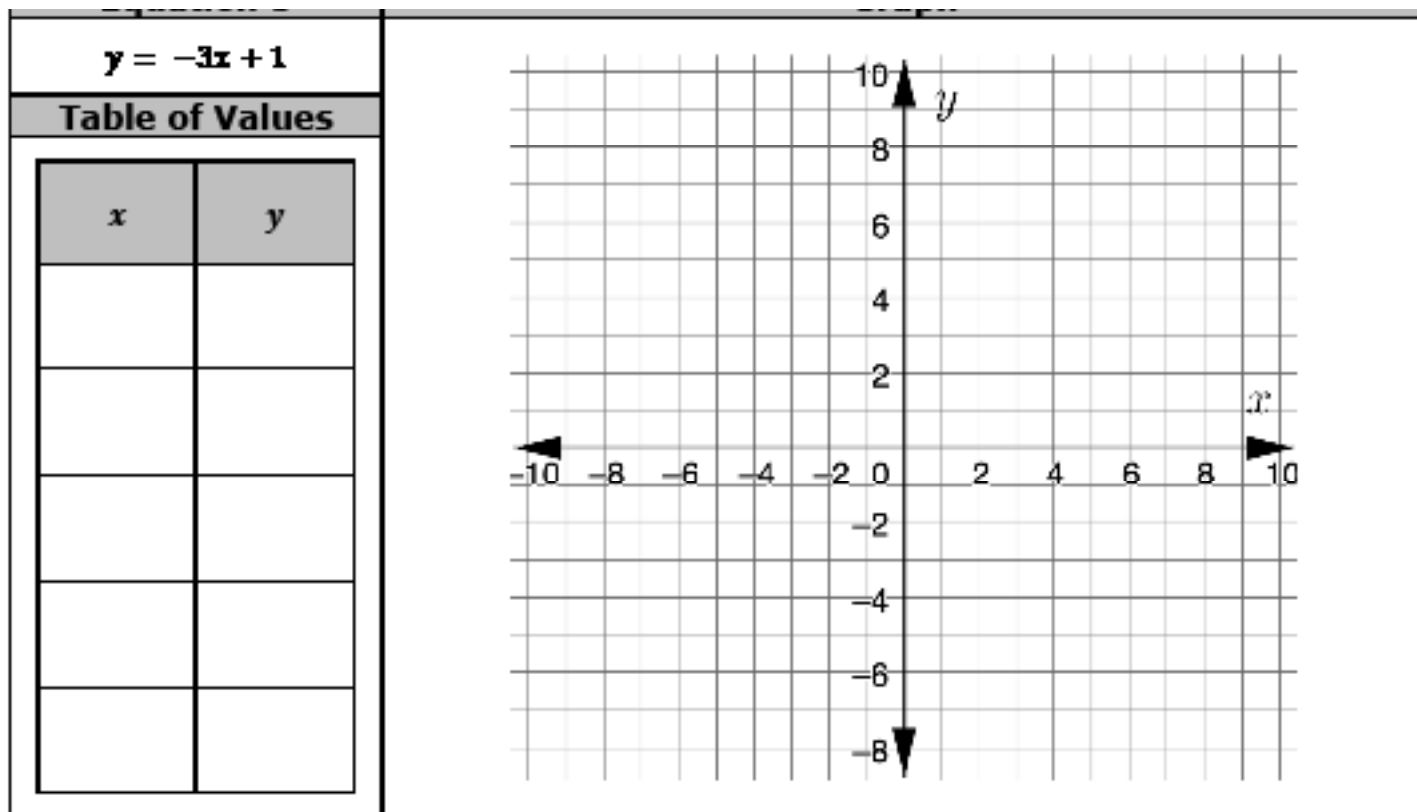
$y = 2^x$

Table of Values

x	y

Graph





2. Use the table below to record your answers to Parts A and B based on the exploration.

Equation 1 $y = 2x - 7$	Equation 2 $y = x^2 + 2$	Equation 3 $y = 2^x$
Equation 4 $y = 4$	Equation 5 $y = x^3$	Equation 6 $y = -3x + 1$

- a. Put an asterisk (*) next to each equation that corresponds to a linear graph.
- b. Circle each equation that corresponds to a graph that is NOT linear.

- 3. Compare the equations that are linear to those that are not linear.**

What do the equations that are linear have in common?	What do the equations that are NOT linear have in common?

- 4. Make a conjecture about the types of equations that are not linear.**

5. **Ask two classmates their conjectures. Record their conjectures and write your summary of their reasons. *You are the only person who should write in the first two columns of the table.* Have the person initial next to your summary stating that your summary was correct.**

Conjectures	My Summary of Their Reasons	Initials

10.8.4 Wrap-Up: Cool Down

1. Eleven equations are shown. Based on your conjecture about the types of equations that are not linear, circle the equations you *predict* will produce a graph that is **not** a line. Use your best judgement. You will have an opportunity to make revisions during the next lesson.

$$y = -2$$

$$y = \frac{1}{2}\sqrt{x} - 1$$

$$\frac{x^3}{z} = y$$

$$y = -x^2 + 6$$

$$5x = 6 + 3y$$

$$x = 5$$

$$y = \frac{2}{5}x - 7$$

$$x = y^2$$

$$y = 3^x - 2$$

$$y = \frac{z}{x} + 1$$

$$y = |x| - 5$$